

HPS: Cortex-A9 x2 @800 MHz [S]

user: tokenizer, Laya runtime (op. lain)

adapter: quant X per-token (int16),

dequant $y = y_{32} \cdot 2^{10} \cdot s_x \cdot s_w$

kernel: laya_mm.ko (TCB)

CMA buffer, alamat fisik, cache sync,

WIN lock, REQ_ID, timeout, IRQ

user tidak pernah memberi alamat fisik

DDR3 1 GB, 32-bit (bersama) [S]

Linux + runtime + model lain

Jendela CMA [WIN_LO, WIN_HI)

W: $44 \times [2304, 768] \text{ int16} = 155,7 \text{ MB}$

(load sekali, cold start)

X: $[L, 768] \text{ int16} \leq 1,5 \text{ MB}$

Y: $[L, 2304] \text{ int32} \leq 9,4 \text{ MB}$

semua base selaras 64 B

FPGA fabric: Imm_top (100 MHz [A])

CSR + validator + watchdog + IRQ

Imm_core: rd master, loader, xbuf/wbuf,
PE 4x8 (dot4 int16), drain, obuf, wr master

64 DSP (sum-of-2 + accumulate)

$\approx 254\text{--}290 \text{ M10K}$, $\approx 4,7\text{k FF}$ [E]

F2S port0 Avalon 64b RO: X, W

F2S port1 Avalon 64b: Y + fence

LW H2F AXI 32-bit $\rightarrow$ CSR (kontrol saja)

f2h_irq0: DONE / ERROR